

Every process function lifts to a unitary process
Draft note. Foundations of Physics style. Not a submission.

Abstract. A process function is a classical map from the outputs of several parties to their inputs such that every local deterministic intervention has exactly one fixed point. Baumeler, Costa, Ralph, Wolf and Zych showed that every such map extends to a bijection by adding a source and a sink, and left open whether the bijection quantizes to a valid unitary process. Daher Ahmed and Kunjwal conjectured that it does. This note argues that it does. The dilation is the permutation unitary of the source-and-sink extension. A process function cannot have two distinct strings that agree on the process output wherever they disagree as strings. That pair is exactly the off-diagonal the contraction needs to kill. Local instruments then induce a CPTP map by Stinespring. Consistency, in the fixed-point sense, is already a quantization condition. It does not have to be imposed again as a measure-zero axiom.

1. The philosophical problem

Backward causation has never been short of paradoxes. It has been short of a criterion that says which loops are admissible without putting the answer in by hand. Black's bilking argument, the grandfather paradox, and the bootstrap are the standard cases. Lewis's contextual can, Novikov's self-consistency principle, and the Greenberger-Svozil cancellation all exclude the bad case. None derives the exclusion from the structure of the loop. The process-function clause does. No fixed point is the grandfather case. More than one is the bootstrap. Dourdent et al. (2026) say the same thing as a complete pairwise-exclusive event list. Cyclic causation is allowed. Cyclic signaling is not. The Lugano function is the witness. The twelve binary two-party process functions are all causally ordered.

2. Why quantization is the next question

Deutsch's density-operator fixed point always exists in finite dimension and is not diagnostic of a loop. Post-selection simulates the event list and is not an embedding. The good demand is unitarity of a dilation. The 2019 extension is $i = w(o) + e$, $s = o$. A bijection is a permutation unitary. The open question is whether parties may insert arbitrary quantum instruments and still induce a CPTP map from source to sink. If the lift holds, the fixed-point clause was already the quantum condition. The source and the sink are the boundary of the loop. Reversibility is the inverse reading, not the demand that w be an involution on its own alphabet.

3. Time symmetry, and what this does not settle

Price, and Leifer and Pusey, argue that a time-symmetric ontology that keeps lambda-mediation cannot refuse influences depending on later settings, because sequential measurements violate a timelike Bell inequality. This lift is adjacent to that no-go, not a solution of it.

A causal-inequality-violating process function already has each input depending on another output. The dilation keeps that dependence. It does not embed the process in a spacetime. The wormhole billiard has data of multiplicity other than one. A process function needs multiplicity one for every intervention. Freedom is also not settled: unique completion is not openness of the choice.

4. The lemma

If w is a process function and s differs from s' , some party k has s_k different from s'_k and $w_k(s) = w_k(s')$. Otherwise a local function can be defined that fixes both strings. A process function cannot have two fixed points. The forbidden pair is the pair that would make the quantum contraction fail.

5. The lift

$U|o, e\rangle = |w(o)+e, o\rangle$. For local unitaries, possibly on an ancilla, the source-to-sink operator is unitary. Off the diagonal the lemma supplies a party at which the shift vanishes and the sink bits differ, so that factor is an inner product of distinct rows and vanishes. Instruments follow by Stinespring. The swap and the identity loop fail the lemma, and their dilations are not trace-preserving.

6. What is interesting if the lemma stands

The reason the function lifts is the same reason it was consistent. The forbidden double fixed point is the forbidden off-diagonal. Only consistent loops occur can mean a filter on solutions, a post-selection, or a definition of the admitted set that is already closed under quantum instruments. The third reading is the one this argument supports.

7. What the argument does not do

It does not embed the process in a spacetime. It does not say the bare function is an involution. It is not kernel-checked. The combinatorial lemma is the piece an attacker has to break.

References

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Repository: github.com/drewarrowood/process-function-unitary-lift